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Early and late Paleozoic remagnetization of the Cambrian Peerless Formation, Colorado

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The Upper Cambrian Peerless Formation in central Colorado contains two secondary components of magnetization. The dominant component was removed during initial AF demagnetization of specimens whereas a weaker component was removed during subsequent thermal demagnetization. IRM acquisition experiments suggest that the dominant component has low coercivity whereas the weaker component has high coercivity. The latter component has southeasterly and shallow directions and the corresponding pole position is located at 40.0°N, 134.5°E. The high coercivity, blocking temperatures over 600°C, and petrographic evidence suggest this component resides in diagenetic hematite. The magnetization is interpreted as a CRM acquired in the late Paleozoic. The low coercivity component directions fall on a great circle that passes through the modern field direction and two modes at 107°E, +10° and 287°E, −10°. This component is interpreted as a vector sum of two antipodal CRMs and a modern VRM. The pole position (10°N, 150°E) suggests acquisition in the early Paleozoic and the remanence resides either in maghemite or in diagenetic magnetite.

1. Introduction

Chemical Remanent Magnetizations (CRMs) are common in sedimentary rocks and can form by the precipitation of authigenic magnetic minerals such as magnetite, hematite and goethite. Some CRMs are all of one polarity, are well grouped, and probably represent acquisition during one episode, perhaps as a result of a specific diagenetic event (Gillett, 1983; Elmore et al., 1985; Loucks and Elmore, 1986). Other CRMs have streaked distributions with both normal and reversed directions. The origin of these streaked magnetizations is not always straightforward. For example, they may represent acquisition during apparent polar wandering over an extended period of time, perhaps as a result of multiple diagenetic events, or they may record a polarity transition (e.g., Helsley and Herrero-Bervera, 1985). Alternatively, a streaked distribution of directions may represent the vector sum of two or more components.

In this paper we report paleomagnetic results from the Upper Cambrian Peerless Formation in central Colorado. The unit contains a complex magnetization, dominated by several CRMs.

2. Geologic setting

The study area is located in central Colorado, west of Manitou Springs (Fig. 1). At this location, the lower Paleozoic stratigraphy begins with the Upper Cambrian (Dresbachian) Sawatch Quartzite, a white to pink sandstone that unconformably overlies the Precambrian Pikes Peak Granite (Fig. 2). The Peerless Formation overlies the Sawatch and is unconformably overlain by the Ordovician Manitou Dolomite. Although the Peerless Formation ranges from Franconian to Trempeleauan in Colorado, according to Berg and

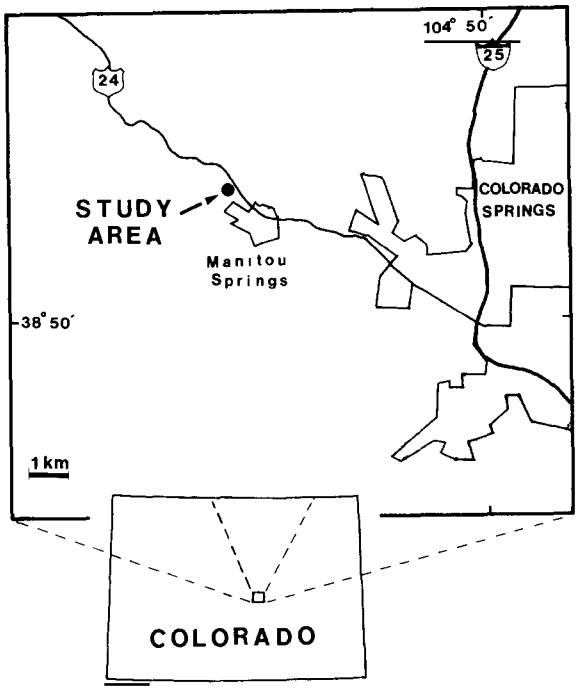


Fig. 1. Location map showing study area.

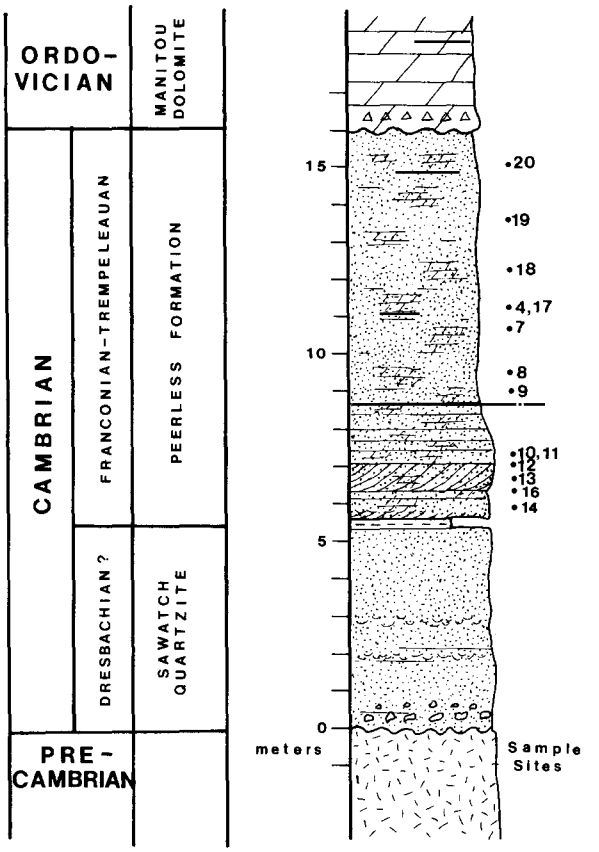


Fig. 2. Schematic stratigraphic section at sampling location. Sample sites are designated by numbers on the right.

Ross (1959) the unit is Franconian in the Manitou Springs area.

The Peerless Formation is a glauconitic dolomitic sandstone at the base that grades into a sandy dolomite at the top of the unit. Cross bedding and other sedimentary structures in the unit indicate deposition in a shallow marine setting, perhaps a tidal flat as Bush (1973) suggested. There has been no previous paleomagnetic work on the Peerless Formation, although Howell and Martinez (1957) reported uncleaned late Paleozoic directions from the Sawatch Formation in Colorado.

The tectonic history in the study area is marked by three periods of activity. During the Ordovician, and continuing through the Devonian, gentle upwarping occurred at least three times with little associated deformation (Ross and Tweto, 1980). In the late Paleozoic, uplift of the Ancestral Rockies apparently did not deform sediments in the sampling area, but sediments derived from the uplift were deposited in the study area. The strata in the sample location were deformed during the Laramide Orogeny and as a result dip to the south.

2. Methods

Oriented samples for paleomagnetic analysis were collected from 16 sites in one gulley about 0.8 km west of Manitou Springs (Figs. 1 and 2). Cores (2.54 cm in diameter) were drilled with a portable gasoline powered drilling apparatus and oriented in situ with a magnetic compass and inclinometer. Specimens from individual sites were collected from single beds of one lithology. The cores were cut to 2.2 cm in the laboratory and the natural remanent magnetization (NRM) of specimens was measured with an ScT 3-axis cryogenic magnetometer. Specimens were demagnetized in a

stepwise fashion using a combination of alternating field (AF) and thermal techniques. Some specimens were thermally demagnetized in 12 incremental steps up to 700°C in a Schonsted TSD-1 unit. Other specimens were demagnetized in ~ 12 steps up to field strengths of 150 mT in a 2-axis tumbling AF unit. In most cases specimens in this group were subsequently subjected to stepwise thermal demagnetization. As will be discussed below, the method involving both demagnetization techniques revealed that many specimens contained two components of magnetization. As a result, this approach was used in most cases.

Directional and intensity results for each specimen were plotted on orthogonal projections (Zijderveld, 1967) and the direction and magnitude of components removed during demagnetization was obtained through visual inspection followed by vector subtraction. Three or more colinear points along a demagnetization path were required to calculate a vector removed from a single specimen.

Isothermal Remanent Magnetization (IRM) acquisition and decay experiments were used to help determine the magnetic mineralogies. Specimens were given a stepwise IRM at increments up to 0.7 T and thermally demagnetized at 50° intervals. Petrographic techniques were also used to help identify the mineralogic carrier of the remanence. Polished thin sections of samples were examined using light microscopy (with transmitted and reflected light). Samples were also examined using a scanning electron microscope (SEM) with an energy dispersive X-ray attachment.

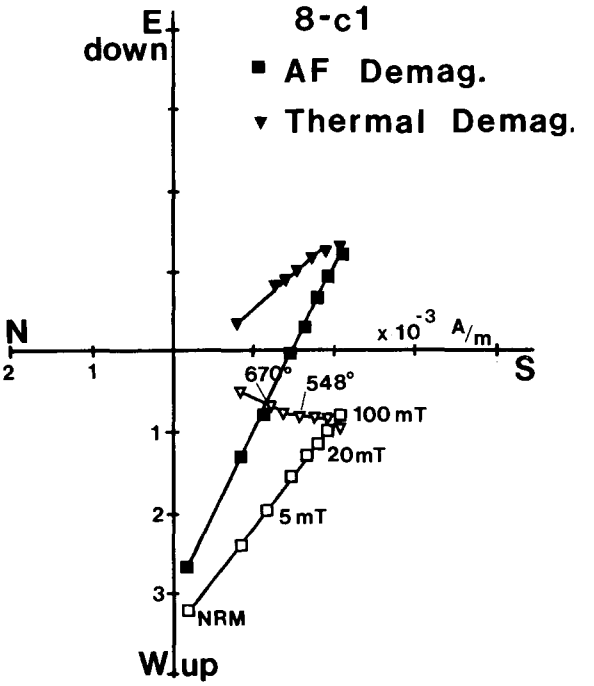


Fig. 3. Orthogonal projection (after structural correction) showing two components of magnetization. One component is removed by AF demagnetization (designated by squares) and a second component is removed by thermal demagnetization (triangles). Open symbols represent projections on the vertical plane and closed symbols represent projections on the horizontal plane. The component removed by AF demagnetization is westerly with steep negative inclination, whereas the component removed during thermal demagnetization is southeasterly with shallow negative inclination.

4. Results

At least two components of magnetization are present within the Peerless Formation. One component was removed by AF demagnetization whereas the other was removed at high temperature by thermal demagnetization (Fig. 3). The component removed by AF demagnetization was present in 70% of the specimens examined and in all specimens which had strong NRM intensities (commonly 1.0×10^{-2} A m^{-1} or greater). Commonly, over 90% of the component was removed by demagnetization to field strengths of 30 mT (Fig. 4a). A few specimens with strong NRM intensities were thermally demagnetized without prior AF treatment and contain a similar direction to that removed by AF demagnetization. Most of the magnetization in these specimens was removed by approximately 525°C (Fig. 4b).

IRM acquisition experiments suggest that the component removed by AF demagnetization is carried by a mineral with low coercivities. The steep initial rise in the acquisition curve (Fig. 4c) suggests a low coercivity phase whereas the increase in intensity above 0.3 T suggests that a higher coercivity phase is also present. Thermal decay of the IRM reveals a range of unblocking temperatures to above 600°C.

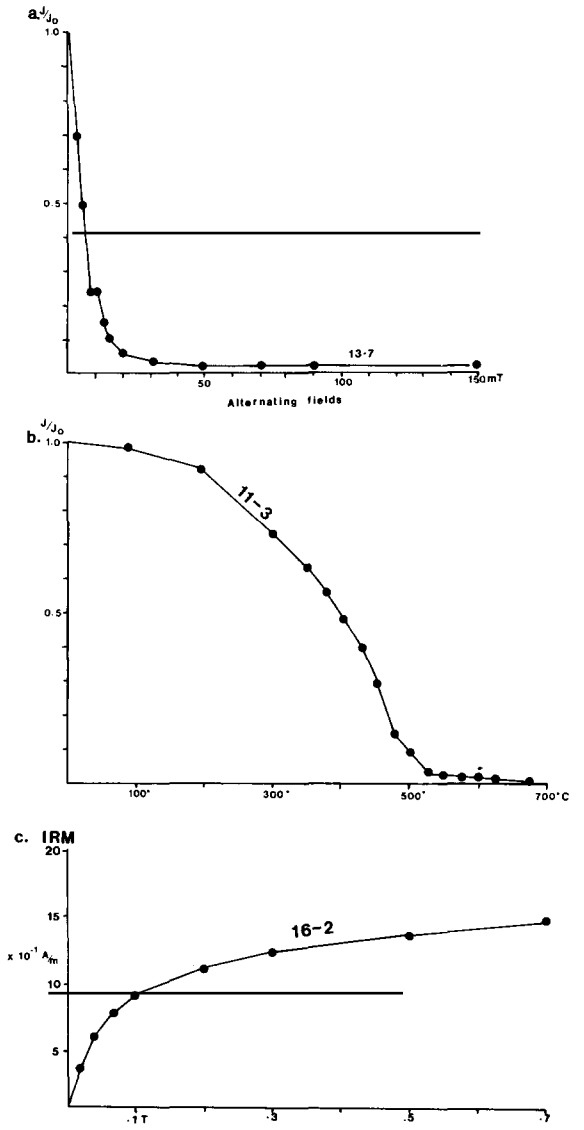


Fig. 4. Demagnetization data for the low coercivity component: (a) J/J_o plot for a representative specimen which shows that 97% of NRM intensity is removed by field strengths of 30 mT. (b) J/J_o plot for thermal demagnetization. Over 96% of the NRM intensity was removed by thermal decay to 525°C. (c) IRM acquisition of a specimen dominated by a low coercivity component. This shows a rapid increase in intensity at IRM field strengths less than 0.1 T followed by a gradual rise.

Specimen directions for this component are streaked, and fall along an inclined great circle that passes through two antipodal modes at

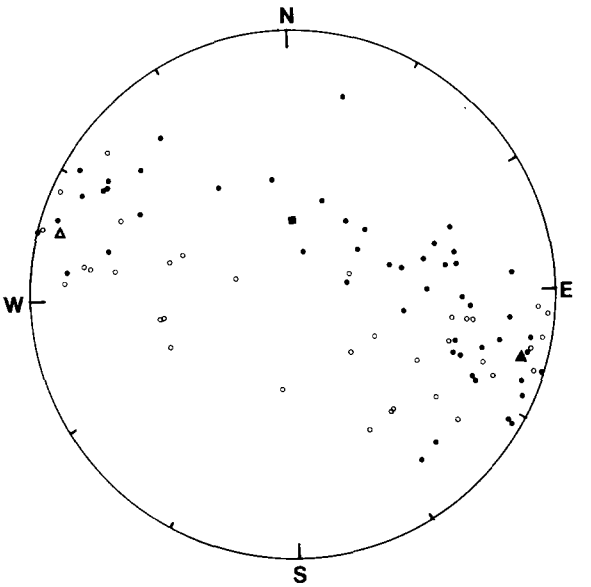


Fig. 5. Specimen directions (after tilt correction) for the low coercivity component plotted on an equal area net (down [+] vectors are solid, up [−] vectors are open). The square is the modern field direction (axial geocentric dipole model) and the two modes are denoted by triangles.

N107°E, +10°, N287°E, −10° and the modern direction (Fig. 5). The modes were calculated using a low pass 3 by 3 matrix filter which smoothed the data and removed outliers. The technique to calculate the modes is similar to the one suggested by Van Alstine (1980). The $\beta 95$ for the southeasterly direction is 9.3° and is 11.5° for the northwesterly direction (Table I). A mean calculated using Fisher (1953) statistics for all the specimens (after reversal of all NW and SW directions) is 105.5°E and +12.6° (Table I).

The low coercivity component was present to some extent in all sites examined; however, the component was more dominant in the lower part of the stratigraphic section. The component decreased in both intensity and occurrence upsection. For example, high in the stratigraphic section (sites 4, 7, 17–20; Fig. 2) the N/No (Table I) is generally lower than for sites low in the section (sites 8–14, 16).

The component removed at high temperatures usually had an NRM intensity in the range of $1.0–3.0 \times 10^{-3}$ A m^{-1}. The specimens were ther-

TABLE I

Data for low coercivity component

Site	*N/No*	Dec.	Inc.	*k*	$\frac{\alpha 95}{(\beta 95)}$	Pole lat.	Pole long.
Unit weighted mean							
all	81/116	105.5	+12.6	4.1	9.0	7.9°N	150.2°E
Modes							
	40/43	107	+10	6.9	9.3	10.0°N	150.0°E
	27/38	287	−10	6.8	11.5	10.0°N	150.0°E
Running means							
A	20/20	74.9	+59.0	24.1	6.8	33.8°S	138.4°E
B	20/20	91.7	+43.5	34.0	5.7	14.2°S	144.1°E
C	20/20	100.4	+28.5	33.3	5.7	1.7°S	152.8°E
D	20/20	106.2	+12.1	32.6	5.8	8.6°N	150.1°E
E	20/20	108.8	−4.4	35.4	5.6	15.9°N	154.8°E
F	20/20	112.2	−20.9	29.6	6.1	24.0°N	159.6°E
G	20/20	126.6	−35.9	16.6	8.3	40.5°N	158.5°E

N/No is the ratio of the number of specimens used in the final analysis to the number of specimens examined. Dec. is magnetic field declination. Inc. is the field inclination. *k* is a precision parameter, and $\alpha 95$ is the radius of the cone of 95% confidence around the mean (Fisher, 1953). Pole lat. is the latitude, and pole long. is the longitude of the pole position corresponding to the declination and inclination for the latitude and longitude of the sample site (lat = 38.87 N, long = 104.92 W). All values are corrected for bedding error (bed strike = 112°, dip = 19° south). The *N/No* for each site: 4, 1/4; 7, 0/5; 8, 11/12; 9, 10/11; 10, 7/9; 11, 12/14; 12, 8/16; 13, 9/10; 14, 6/10; 16, 4/4; 17, 3/4; 18, 1/3; 19, 5/7; 20, 4/7.

mally demagnetized after prior AF treatment to 150 mT and typically have most unblocking temperatures above 600°C (Fig. 6a). Specimens from high in the stratigraphic section (e.g., sites 4 and 7) had little, if any, low coercivity component. IRM acquisition and decay curves for these specimens suggest that the magnetization is carried by a high coercivity mineral with maximum unblocking temperatures above 600°C (Fig. 6b, c). As previously stated, IRM acquisition curves for specimens dominated by the low coercivity phase also indicate the presence of a high coercivity phase (Fig. 4c).

The directions for the high temperature component are generally southeasterly and shallow, with some northwesterly directions also present (Fig. 7). The mean is located at 138.5°, −11.3 (Table II). Examination of a projection with the mean of all directions at the center indicates that the data distribution is close to Fisherian with only slight streaking.

Although there was evidence that the high temperature component occurs in almost all specimens examined, the component could not be adequately identified in many specimens. A total of 118 specimens were examined for the high temperature component, although only 69 specimens were used in statistical analysis. Specimens were omitted for several reasons. Some specimens (44) had magnetizations that were southeasterly or northwest-

TABLE II

Data for high temperature component. See Table I for explanations

Site	*N/No*	Dec.	Inc.	*k*	$\alpha 95$	Pole lat.	Pole long.
Unit weighted mean							
all	69/118	138.5	−11.3	17.6	4.1	40.0°N	134.5°E

N/No ratios for each site: 4, 6/10; 7, 8/10; 8, 12/12; 9, 3/11; 10, 0/9; 11, 4/14; 12, 5/12; 13, 6/8; 14, 9/10; 16, 0/4; 17, 1/3; 18, 2/2; 19, 7/7; 20 6/6.

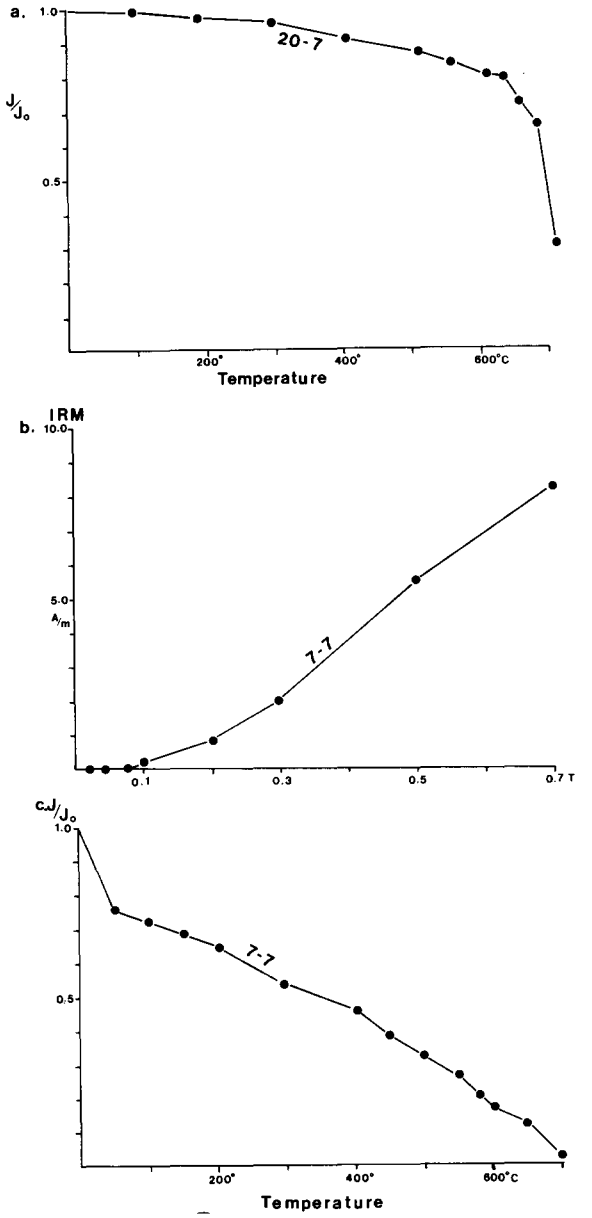


Fig. 6. Demagnetization and rock magnetic data for the high temperature, high coercivity component: (a) J/J_o for thermal demagnetization. Note the range of unblocking temperatures with most above 600°C. (b) IRM acquisition for a specimen with no low coercivity component. The lack of a rapid increase of intensity at field strengths below 0.3 T shows that the specimen only contains a component with high coercivities. (c) Thermal decay of IRM shows unblocking temperatures in part greater than 600°C.

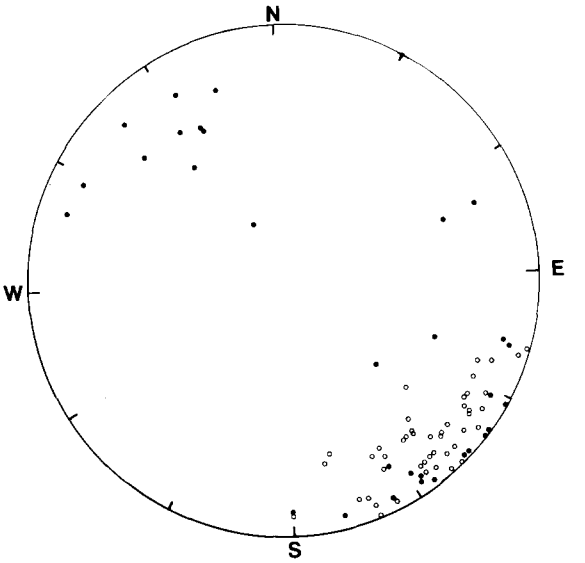


Fig. 7. Directional data (after tilt correction) of the high temperature, high coercivity component.

erly but they did not contain any single demagnetization path that could be isolated. Other specimens (5) had present-day field or other directions.

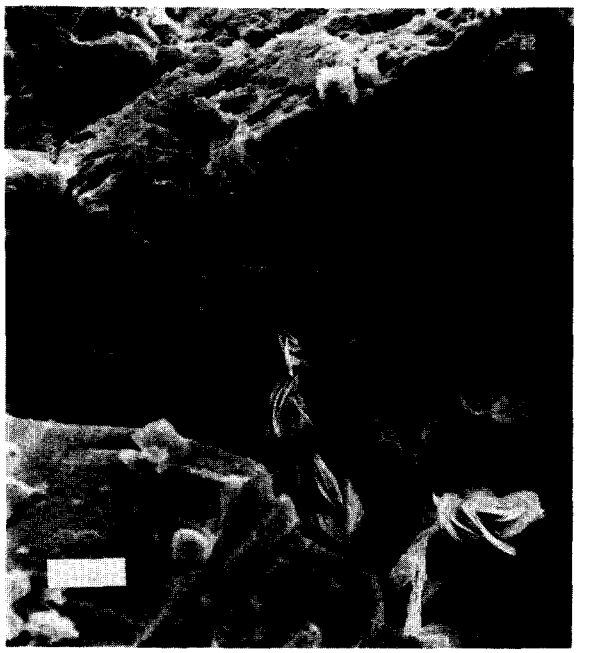


Fig. 8. SEM photomicrograph illustrating authigenic hematite associated with dolomite. Bar scale = 10 μm.

Authigenic hematite is common in the samples of the Peerless Formation. For example, authigenic hematite is particularly common within and on dolomite rhombs (Fig. 8). Hematite also occurs replacing glauconite grains, as accumulations in pore spaces between detrital and diagenetic phases, associated with calcite that is replacing silicate grains (quartz or feldspar) and in association with quartz overgrowths.

5. Interpretation of results and discussion

As previously stated the directions for the low coercivity component are streaked and fall on a great circle. There are several explanations for the distribution of the data. First, the streaked pattern could represent acquisition of magnetization over an extended time interval during apparent polar wander. To evaluate this issue pole positions were calculated for the streaked path using a sliding running mean (Table I). The poles fall along a path between 34°S, 138°E and 41°N, 159°E. Some of the running-mean poles fall close to Cambro–Ordovician poles from other units, although most of the running mean poles fall to the NE or SW of any generally accepted portion of the Late Cambrian and younger APWP.

A second and more likely possibility is that the inclined great circle distribution represents the vector sum of normal, reversed, and modern directions. As previously stated, the great circle distribution appears to intersect the modern direction (Fig. 5). The fact that some southerly and steep negative directions are present suggests that some specimens may also contain a reversed modern component. Modern viscous remanent magnetizations (VRM) are commonly removed at low field strengths during AF demagnetization. The low coercivity magnetization was removed at low field strengths, which supports the suggestion that a modern VRM may be one unresolved component. To investigate whether the data distribution could be due to the mixing of these components we mixed two nearly antipodal directions and the modern direction in various proportions. A streaked path very similar to that observed for Peerless specimens was produced, and supports our interpretation that the great circle distribution is a vector sum of those components. The streaked distribution could not result from the addition of two antipodal components alone, because the vector sum of two antipodal components produces a large amount of scatter, especially with approximately equal amounts of each.

The magnetic mineralogy of the low coercivity component is uncertain; it could be magnetite, maghemite, or low coercivity hematite. The NRM intensities of specimens with the low coercivity component are relatively strong (commonly $1.0–2.0 \times 10^{-2}$ A m^{-1}), which suggests a high saturation magnetization. This and the low coercivity suggest that the remanence is not carried in low coercivity hematite. The component probably resides in magnetite or maghemite. Maghemite has the chemistry of hematite and the crystallographic and magnetic properties of magnetite and can form by low temperature oxidation of magnetite. The rock magnetic results are consistent with either mineral. Neither maghemite or magnetite were found during the petrographic study; hematite was the only magnetic mineral found.

The pole position for the low coercivity component is 150°E, 10°N and is near poles thought to be Cambrian, as well as Ordovician in age (Fig. 9). Because of uncertainties about Cambrian APW, all that can reliably be inferred is that the magnetization is probably early Paleozoic in age. The pole position, however, does support other work which suggests low-latitude pole positions for Late Cambrian rocks (see Fig. 9). This is in contrast with a recently suggested lower Paleozoic apparent polar wander path for North America by Larson et al. (1985) which has Late Cambrian poles at latitudes greater than 20°N.

The low coercivity component contains both normal and reversed directions within single sample sites, and as no vertical pattern of reversals is observed, the magnetization is suggested not to be a DRM. The high unblocking temperatures reached during thermal demagnetization (> 525°C) suggest relatively high burial temperatures based on relaxation time-blocking temperature curves for magnetite (Middleton and Schmidt, 1982). This, and the early Paleozoic age of the magnetization suggest that a thermoviscous origin is not

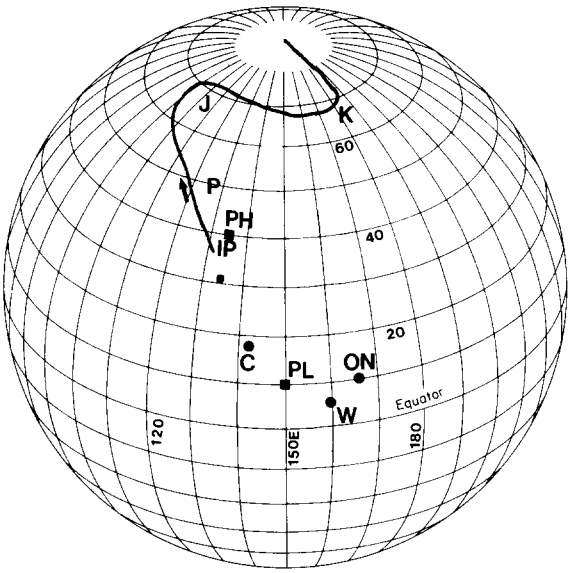


Fig. 9. The Apparent Polar Wander Path for the late Paleozoic through the present (after Irving and Irving, 1982) and selected other pole positions from Cambrian and Ordovician rocks: PL, low coercivity component, this study; PH, high coercivity component, this study; W, Wilberns Formation (Van der Voo and others, 1976); ON, Oneota Formation (Jackson and Van der Voo, 1985); and C, Colorado Intrusions (Lynnes and Van der Voo, 1984).

likely. Although the evidence is largely negative, the component is interpreted as a CRM.

Unblocking temperatures greater than 580°C and the high coercivities suggest that the second component resides in hematite. The pole for this component (40° N, 134° E) falls near the late Paleozoic APWP (Fig. 9). Because the Peerless is Franconian, the pole should not be compared with other similar poles that are interpreted as Dresbachian and were used to construct the Cambrian loop (Watts et al., 1981). The component is therefore dominantly late Paleozoic in age.

Because the magnetization is secondary, it is probably of thermoviscous or chemical origin. The high temperatures reached during thermal demagnetization (> 650°C for many specimens) require high burial temperatures (> 550°C) based on the relaxation time-blocking temperature relationships (Pullaiah et al., 1975). There is no evidence that the rocks were heated to such high temperatures; a thermoviscous origin, therefore, is probably not likely. The magnetization is interpreted as a CRM.

The fact that both normal and reversed directions of the component are present and that there is no vertical pattern of reversals is consistent with this interpretation. Additional evidence of a CRM comes from petrographic data indicating extensive diagenetic hematite.

Acknowledgments

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References

Berg, R.W. and Ross Jr., R.J., 1959. Trilobites from the Peerless and Manitou Formations, Colorado. J. Paleontol., 33: 106–119.

Bush, J.H., 1973. The Upper Cambrian Stratigraphy of Central Colorado. Ph. D. thesis, Washington State University, Pullman, 231 pp.

Elmore, R.D., Dunn, W.J. and Peck, C.J., 1985. Absolute dating of a diagenetic event using paleomagnetic techniques. Geology, 13: 558–561.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc., Ser. A., 217: 295–305.

Gillett, S.L., 1983. Major, through-going stylolites in the Lower Ordovician Goodwin Limestone: petrography with dating from paleomagnetism. J. Sediment. Petrol., 53: 209–219.

Helsley, C.E. and Herrero-Bervera, E., 1985. Reply to comments on Paleomagnetism of a polarity transition in the Lower (?) Triassic Chugwater Formation, Wyoming. J. Geophys. Res., 90: 2063–2065.

Howell, L.G. and Martinez, J.D., 1957. Polar movement as indicated by rock magnetism. Geophysics, 22: 384–397.

Irving, E. and Irving, A., 1982. Apparent polar wander paths, Carboniferous through Cenozoic and the assembly of Gondwana. Geophys. Surv., 5: 141–188.

Jackson, M. and Van der Voo, R., 1985. A Lower Ordovician paleomagnetic pole from the Oneota Dolomite, upper Mississippi River Valley. J. Geophys. Res., 90: 10 449–10 461.

Larson, E.E., Paterson, P.E., Curtis, G., Drake, R. and Mutschler, F.E., 1985. Petrologic, Paleomagnetic and structural evidence of a Paleozoic rift system in Oklahoma, New Mexico, Colorado and Utah. Geol. Soc. Am. Bull., 96: 1364–1372.

Loucks, V.L. and Elmore, R.D., 1986. Absolute dating of dedolomitization and origin of magnetization in the Cambrian Morgan Creek Limestone: Geol. Soc. Am. Bull., 97: 486–496.

Lynnes, C.S. and Van der Voo, R., 1984. Paleomagnetism of the Cambro–Ordovician McClure Mountain alkalic complex, Colorado. Earth Planet. Sci. Lett., 71: 163–172.

Middleton, M.F. and Schmidt, P.W., 1982. Paleothermomentry of the Sydney Basin. J. Geophys. Res., 87: 5351–5359.

Pullaiah, G., Irving, G., Buchan, K.L. and Dunlop, D.J., 1975. Magnetization changes caused by burial and uplift. Earth Planet. Sci. Lett., 28: 133–143.

Ross, R.J. and Tweto, O., 1980. Lower Paleozoic sediments and tectonics in Colorado In: H.C. Kent and K.W. Porter (Editors), Colorado Geology. 1980 Symposium Rocky Mountain Association of Geologists, Denver, pp. 47–56.

Van Alstine, D.R., 1980. Analysis of the modes of directional data with particular reference to paleomagnetism. Geophys. J.R. Astron. Soc., 61: 101–113.

Van der Voo, R., French, R.B. and Williams, D.W., 1976. Paleomagnetism of the Wilberns Formation (Texas) and the Late Cambrian paleomagnetic field for North America. J. Geophys. Res., 85: 5316–5330.

Watts, D.R., Van der Voo, R. and Reeve, S.C., 1981. Cambrian paleomagnetism of the Llano uplift, Texas. J. Geophys. Res., 85b: 5316–5330.

Zijderveld, J.D.A., 1967. Demagnetization of rocks: Analysis of results. In: D.W. Collinson et al. (Editors), Methods in Paleomagnetism. Elsevier, New York, pp. 254–286.